ANGLE OF ARRIVAL ESTIMATION VIA PHASE DIFFERENCE ACROSS TWO SDRS

Individual Course of Study Final Report

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Abstract

This project developed and validated a dual-bladeRF software-defined radio (SDR) system for phase-based angle-of-arrival (AoA) estimation. The system integrates receiver synchronisation, RF switching, automatic calibration, continuous signal correction and AoA estimation into a complete processing pipeline. Calibration algorithms were validated through simulation and hardware testing, demonstrating effective compensation of relative time-delay, amplitude and phase differences between receivers. Continuous dual-SDR experiments achieved mean corrected correlations exceeding 0.99, with sample alignment maintained over extended acquisitions.

The principal limitation identified was an inconsistent residual phase offset when transferring calibration between the RF switching board's NOISE and ANTENNA paths. Ten independent system starts produced a total phase-offset spread of 15.084°, despite relatively stable phase measurements within individual acquisitions. Further investigations confirmed that the variation persisted when using an independent external RF source, although its underlying cause remained unresolved. Simulated AoA estimation achieved an angular RMSE of 0.070° at 30° arrival, 10 dB SNR and 10,000 samples. Introducing an unknown phase offset uniformly distributed over ±7.5° increased RMSE to 1.477° under a separate 5,000-sample test, demonstrating the importance of accurate phase calibration.

The final integrated hardware experiment successfully processed 18,000 consecutive measurement windows, achieving a mean corrected correlation of 0.9740 and an apparent AoA standard deviation of 0.0381°. As the experiment used a common conducted RF signal rather than spatially separated antennas, absolute angular accuracy was not established. The majority of project requirements were met, with R3.2 and R4.6 partially satisfied and R6 demonstrated through simulation only. The project demonstrates a functional dual-SDR AoA estimation pipeline, with the primary remaining challenges being resolution of the startup-dependent phase uncertainty and validation against known physical arrival angles.

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1. Introduction

Software-defined radios (SDRs) provide a flexible platform for the acquisition and digital processing of radio-frequency (RF) signals, allowing signal-processing techniques to be implemented and evaluated without requiring dedicated hardware for each application. This flexibility makes SDRs particularly useful for applications such as direction finding and angle-of-arrival (AoA) estimation, where the relative characteristics of signals received at multiple channels can be used to determine the direction of a transmitting source.

AoA estimation relies on accurately determining the differences between signals received at spatially separated antennas. In particular, the relative phase and time delay between the received signals provide information about the difference in propagation distance to each antenna. However, these measurements can be affected by differences in the signal paths and hardware used by each receiver. Relative channel delays, phase offsets and amplitude differences can introduce errors that are indistinguishable from changes caused by the propagation environment. Accurate synchronisation and calibration of the receiving channels are therefore important prerequisites for reliable AoA estimation.

This report investigates the development of a dual-BladeRF system for AoA estimation, covering the synchronisation, calibration and continuous processing of signals received by two independent SDRs. The work includes the development and refinement of algorithms for correcting relative time delay, amplitude and phase differences, alongside the integration of RF switching hardware to support calibration and subsequent signal reception. AoA estimation is evaluated through simulation before being incorporated into the complete hardware processing pipeline. Experimental testing assesses system performance and repeatability, including the investigation of residual phase offsets that remain a limitation for accurate hardware AoA estimation.

The remainder of this report describes the theoretical basis of the methods used, their implementation and experimental evaluation, and the design challenges and performance of the resulting system.

2. Context

2.1 Problem Definition

The project aims to develop a two-channel BladeRF SDR system for AoA estimation using the relative phase of received signals. To achieve this, the system must account for differences between the two receiver channels that affect the measured amplitude, phase and timing of the received signals. The project therefore involves:

- Characterising the relative amplitude, phase and time-delay offsets between BladeRF receiver channels.
- Developing a calibration algorithm to estimate and compensate for these channel-dependent offsets.
- Establishing synchronisation and phase coherence between two BladeRF SDRs.
- Developing an AoA estimation algorithm using the calibrated relative phase between the receiver channels.

Project Scope: The report covers both initial calibration algorithm development and the two-SDR system. Including a common-clock operation, continuous IQ acquisition, calibration correction and optimisation, RF switching investigations, and a one-dimensional AoA estimator. The AoA algorithm has been validated with simulated signals however absolute direction measurements using physical antennas remain outside the demonstrated results.

2.2 Objectives and Requirements

Objective - Develop a system using two BladeRF transceivers to characterise the relative timing, amplitude and phase of received RF signals and use the calibrated phase difference to estimate their AoA.

The project was divided into two main stages, with the first focusing on establishing the synchronisation and calibration required for reliable phase measurements and the second focused on the AoA estimation. A full analysis against the requirements can be found in Appendix A.

Stage 1 - SDR synchronisation/phase coherence

Table 1: Full list of requirements defined at the start of the project.

Number	Requirement	Acceptance Criteria
R1	The system shall receive RF signals from two synchronized BladeRF transceivers.	Samples shall be successfully acquired from both SDR channels using a common reference.
R2	The system shall characterise the relative differences between the two SDR receiver channels.	The relative amplitude, phase, and time-delay differences between channels shall be estimated.
R2.1	The system estimates the relative amplitude difference between the two SDR receiver channels.	Yes/No
R2.2	The system estimates the relative phase difference between the two SDR receiver channels.	Yes/No
R2.3	The system estimates the relative time delay between the two SDR receiver channels.	Yes/No
R3	The system shall calibrate the SDR receiver channels to compensate for hardware-induced offsets.	The calibration procedure shall use the estimated channel differences and demonstrate compensation of hardware-induced offsets.
R3.1	The system shall estimate the hardware-induced channel offsets using a common reference signal.	Relative amplitude, phase, and time-delay offsets shall be determined using the calibration reference signal.
R3.2	The system shall compensate for hardware-induced offsets between the SDR receiver channels.	Calibrated measurements shall demonstrate reduced channel mismatch compared with uncalibrated measurements.
R4	The system shall validate the performance and stability of the SDR calibration.	The accuracy, robustness, and stability of the calibration shall be evaluated under varying operating conditions.
R4.1	The system shall evaluate the accuracy of the amplitude, phase, and time-delay estimates.	The estimation error shall be quantified and reported.

R4.2	The system shall characterise the effect of SNR on estimation accuracy.	Estimation accuracy shall be evaluated over a range of SNR values.
R4.3	The system shall characterise the effect of sample count on estimation accuracy.	Estimation accuracy shall be evaluated for varying sample sizes.
R4.5	The system should characterise the temporal stability of the channel calibration.	The change in amplitude, phase, and time-delay offsets shall be evaluated over time under fixed operating conditions.
R4.6	The system shall characterise the stability of the channel calibration under different operating conditions.	The impact of time, SDR configuration changes, and operating parameters on channel calibration shall be evaluated.
Stage 2 - Estimating Angle of Arrival		
R5	The system calculates the angle of arrival of the received waveform	Yes/No
R6	The angle of arrival should be calculated to within a determined tolerance.	Yes/No

2.3 Hardware

The hardware used during development and validation is listed below. Early work used one bladeRF for testing and validation of the calibration algorithm. Later experiments used two bladeRF 2.0 micro devices and a shared 38.4 MHz reference clock.

- Two bladeRF 2.0 micro SDRs – master/slave receiver channels, with master CLKOUT connected to slave CLKIN.
- Coaxial cables and 50 Ω terminations – controlled wired signal paths for calibration and hardware validation.
- RF splitters and attenuators – common test signals distributed safely to both receiver paths.
- RF switching/calibration PCB – selects common NOISE calibration or independent ANTENNA signal paths.
- BG7TBL external broadband noise source – independent common ANTENNA-mode reference for phase-transfer testing.

2.4 Software

Several software tools were used throughout the project for signal generation, acquisition, processing and validation.

- GNU Radio – used for early signal generation, IQ capture and hardware prototype experiments.
- Python with libbladeRF, NumPy and SciPy – used for continuous dual-SDR acquisition, calibration estimation and optimisation, streaming signal correction, logging and AoA estimation.

3. Background Theory/Research

3.1 BladeRF SDR's

The BladeRF is a software-defined radio (SDR) platform that provides configurable transmit and receive channels for the generation, transmission, and acquisition of RF signals. The device

interfaces with a computer, allowing the captured signals to be processed using software rather than dedicated hardware. In this project, the BladeRF is used to generate and acquire controlled RF signals, providing a platform for investigating the relative timing, amplitude and phase characteristics of receiver channels.

The two devices share a 38.4 MHz frequency reference, but start receiving at independently determined sample positions. Software alignment is therefore still necessary; a shared clock alone does not establish simultaneous samples or a repeatable initial RF phase.

3.2 AoA and Relative Phase

Angle-of-arrival estimation determines the direction from which a signal is received. A simple approach uses two spatially separated receiving antennas. When a signal arrives at an angle, the difference in propagation distance produces a relative time delay, appearing as a phase difference for a narrowband signal. Given the antenna spacing and signal wavelength, this measured relative phase can be converted into a direction relative to array broadside.

For two antennas separated by a distance d, the phase difference due to the signal path difference can be expressed as:

$$\Delta\phi = -\frac{2\pi d sin(\theta)}{\lambda}$$

where θ is the angle of arrival and λ is the wavelength of the received signal. Rearranging gives

$$\theta = \sin^{-1}\left(-\frac{\Delta\phi}{2\pi(\frac{d}{\lambda})}\right)$$

In practice, the measured phase difference is not solely determined by the direction of arrival. Differences between the two receiver signal paths can introduce additional phase and timing differences. These channel-dependent differences must therefore be characterised and compensated for, allowing the remaining phase difference to be used to estimate the signal's direction of arrival.

For a two-antenna system, the ratio (d/λ) represents the antenna spacing relative to the signal wavelength. To avoid ambiguity in the estimated arrival angle, this ratio is typically limited to 0.5 or less [1]. AoA estimates also become more sensitive to phase errors when signals arrive nearly parallel to the line connecting the antennas. Additionally, two antennas can only estimate one component of the arrival direction, determining both requires additional antenna measurements.

3.3 Channel-Dependant Offsets

The accuracy of an angle-of-arrival calculation is dependent on the characterisation of the receiving hardware. AoA estimation relies on the phase difference between signals received by two radios separated by a known distance. However, differences between the two RF signal paths can introduce additional phase offsets, time delays and amplitude variations that are unrelated to the direction of arrival. These differences may result from unequal RF cable lengths, connectors, filters, amplifiers, mixers, or other components within the receiver signal chains.

These channel-dependant mismatches introduce differences in amplitude, phase, and time delay between the two signals. They can be represented using the model in Equation 1.

$$\hat{y} = Ax[n-\tau]e^{j\phi} \qquad \text{Eqn 1}$$

Where $x[n]$ is the reference signal, and $\hat{y}$ is the expected signal at the second channel after accounting for the relative time delay τ, phase difference ϕ, and amplitude difference A.

In practice receiver 1 will observe $x[n]$ and receiver 2 will observe $y[n]$. A calibration algorithm can then be used to determine the three relative differences between the signals. Once these offsets have been determined, they can be compensated for in software to produce a calibrated signal whose amplitude, phase, and timing are aligned with the reference channel.

3.4 Delay Estimation

A key component of characterising the channel-dependent effects is determining the relative time delay between the receiver channels. The algorithm investigated and implemented in this report is the delay estimation method developed by Markus Nentwig [2]. The method estimates the relative delay between two signals, along with their complex scaling factor, which can subsequently be used to determine the relative amplitude and phase. The delay estimate is particularly important for this project because it allows the time-delay contribution of the hardware to be quantified and subsequently accounted for.

The method first obtains a coarse timing delay using cross-correlation. The coarse delay is an integer-sample estimate, with the peak of the cross-correlation indicating the relative shift at which the two signals are best aligned. The resolution of this estimate is therefore limited to one sample, with the corresponding time resolution determined by the sampling frequency.

In the original phase-slope method, the remaining fractional-sample delay is estimated from the relative frequency-domain phase of the coarse-aligned signals. Since a time delay produces a linear phase shift with frequency,

$$\phi(f) = -2\pi\tau$$

the slope of the relative phase can be used to estimate the remaining delay. A weighted least-squares fit to the unwrapped relative phase provides the original fine-delay estimate, which is combined with the integer delay. Later experiments found this unwrapping approach could fail on noisy spectra; the final calibration optimiser instead selects a bounded fractional delay by maximising coherent correlation, as described in Section 4.6.

Once the relative delay is estimated, the aligned complex signal is projected onto the reference to obtain a least-squares complex scaling factor. Its magnitude and phase represent the relative amplitude and constant phase offsets. Figure 1 illustrates the original Nentwig method, which provided the foundation for the later optimised implementation.

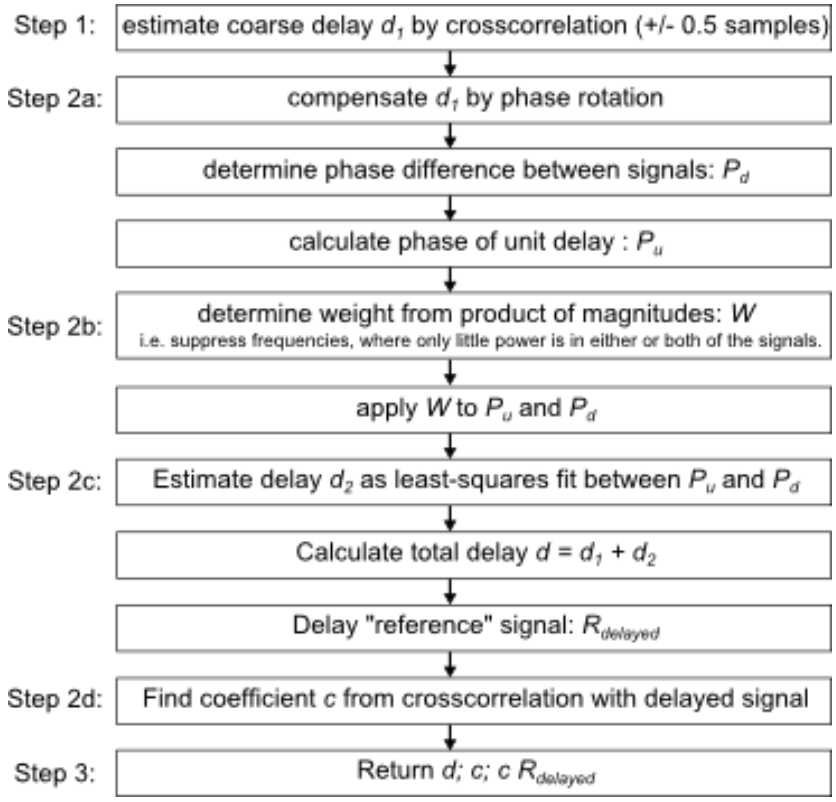


Figure 1: Markus Nentwig delay estimation algorithm process [1].

4. Design, Methodology and Testing

4.1 System Overview

The overall project methodology is shown in Figure 2. The project was divided into four main development branches, covering SDR synchronisation, channel-offset compensation, calibration algorithm development and AoA estimation. Each component was developed and tested individually before integration into the overall system.

The final software system uses two bladeRF receivers sharing a common clock reference. The master receiver provides the reference signal, while the slave receiver is calibrated against it to correct relative timing, amplitude and phase differences. Calibration parameters are estimated and validated using separate sample captures before being applied to continuously acquired IQ samples for AoA estimation.

The following sections describe the development and validation of these components, including the experimental methods used to evaluate their performance. While the calibration system has been validated on hardware and the AoA algorithm has been validated in simulation, final experimental AoA validation using physical antennas has not yet been completed.

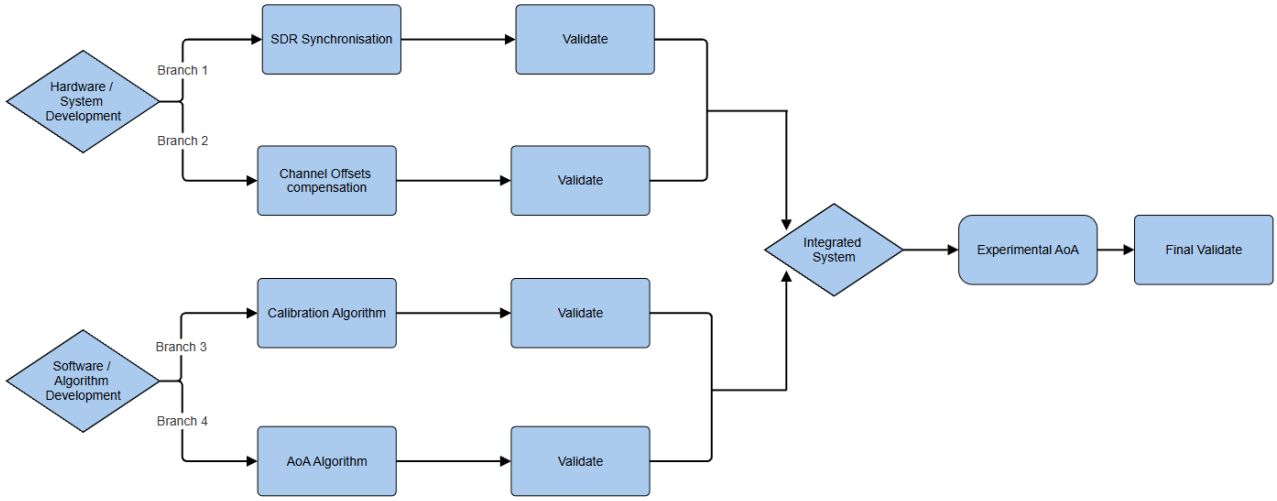


Figure 2: Modular Overview of AoA development pipeline.

4.2 Hardware Configuration and Data Acquisition

4.2.1 Hardware Configuration

To synchronise the two SDRs, they must share a common clock reference. One bladeRF is configured as the master, providing its internal 38.4 MHz reference clock to the slave through the CLKOUT and CLKIN connections [3]. While this ensures that both SDRs share the same reference frequency, it does not guarantee identical receiver phase, sample alignment or repeatable phase after startup.

The hardware setup can be seen in Figure 3, consisting of the following elements:

- **Master SDR (Black) –** Provides the shared reference clock and transmits the noise signal used for calibration through TX0. Its RX0 port is connected to the SDR1 output of the RF switching board.
- **Slave SDR (Red) –** Receives its clock reference from the master through CLKIN. Its RX0 port is connected to the SDR2 output of the RF switching board.
- **RF Switch Board -** Receives the common calibration signal from the master through J2 (NOISE) and routes either this signal or the ANT inputs to the SDR receiver outputs.
- **4-way Splitter –** Divides the external noise signal between the two ANT inputs for calibration-transfer testing.
- **External Noise Source –** Provides an independent broadband noise signal to investigate calibration transfer and relative phase stability.
- **Attenuators –** Located between the external noise source and splitter, providing 32 dB attenuation to reduce the received signal power to a safe level for the SDRs.
- **50Ω Terminators –** Used to terminate unused RF ports, including the unused splitter output.

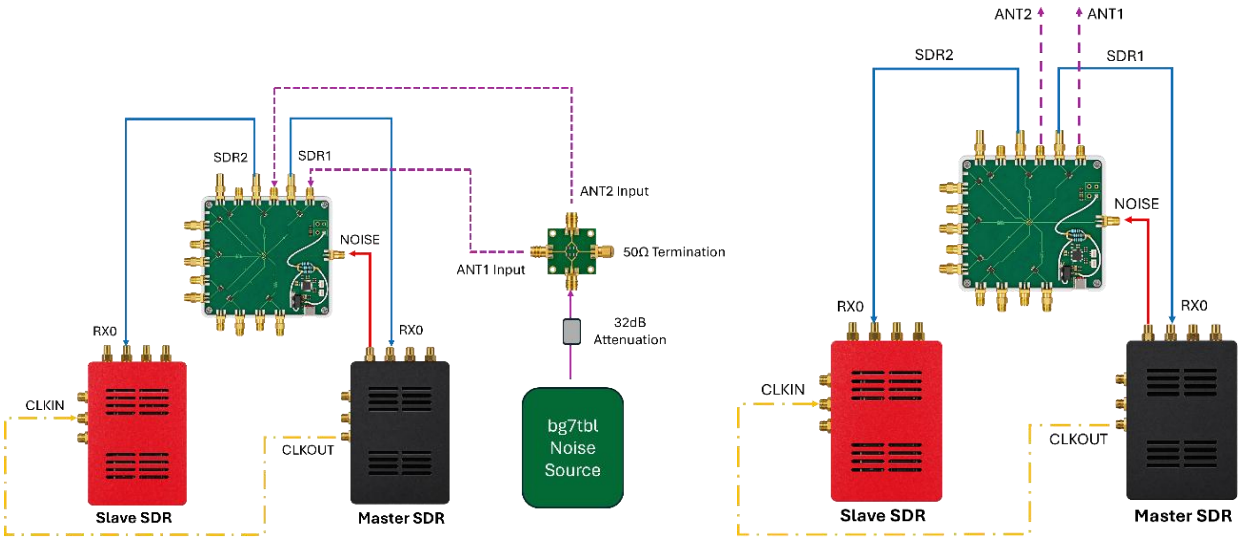


Figure 3: (Left) Testing hardware setup with external noise source. (Right) Recommended applied hardware setup.

The configuration shown on the left of Figure 3 was used for the external noise experiments described in later sections. For physical AoA measurements, the external noise source and splitter are removed, and antennas are connected to the ANT inputs, as shown on the right of Figure 3.

The RF switching board allows the receiver inputs to be switched between a common calibration signal and the antenna inputs without physically reconnecting RF cables. This reduces manual intervention and allows the calibration and measurement process to be automated. However, differences between the RF paths may introduce additional phase offsets that are not accounted for by the initial calibration. The stability and repeatability of these offsets were investigated experimentally.

4.2.2 Software Configuration

The first two stages of the software pipeline initialise the RF switching board and configure the SDRs before calibration begins.

Stage 1 – RF Switch Board Setup

The RF switching board is controlled using the CaliBoard Python module, provided by Adam Smith alongside the hardware. Communication is established through a serial interface, allowing the software to switch between NOISE and ANTENNA modes.

The board uses Skyworks SKY13330-397LF RF switches, which have a reflective configuration [4]. This can result in reflected RF power when a signal is applied to a disconnected path. To avoid switching under active transmission, the board is placed into NOISE mode before the calibration transmitter is enabled. Similarly, transmission is stopped before switching back to ANTENNA mode.

Stage 2 – SDR Setup

The startup.py module identifies the two bladeRF devices using their unique serial numbers and configures their clock sources. The master is configured to use its internal VCTCXO reference and provide a clock output, while the slave is configured to accept this external reference.

Both receiver channels are then configured using the following settings:

- **Centre frequency:** 100 MHz
- **Sample rate:** 1 MS/s

- **Receiver bandwidth:** 750 kHz
- **Receiver gain:** 10 dB, using manual gain control

Following SDR initialisation, the calibration transmitter is configured separately through transmission.py. This uses the master's TX0 channel with a centre frequency of 100 MHz, sample rate of 1 MS/s, bandwidth of 750 kHz and transmitter gain of 30 dB.

Once the hardware has been configured, the calibration noise signal is transmitted and continuous IQ acquisition begins, allowing the relative receiver offsets to be estimated.

4.2.3 Continuous Data Acquisition

Continuous IQ acquisition was implemented using the DualSDRAcquisition Python class. Each bladeRF receiver operates in a separate thread, allowing samples to be acquired continuously and independently of the main processing pipeline. The receivers use the SC16_Q11 format (defined by the BladeRF libbladeRF library [5]), transferring blocks of 8,192 complex samples, which are converted into normalised complex floating-point values.

Received samples are stored in separate queues for the master and slave SDRs. The software retrieves equal-sized blocks from each queue for processing, preserving any unused samples for the following window. This allows the acquisition block size to differ from the processing window size without discarding data. Queue overflow and receiver errors are also monitored to prevent processing from continuing with potentially incomplete data.

Although the SDRs share a clock reference, their acquisition threads are started independently, meaning the resulting data streams are not necessarily aligned in time. The initial sample offset must therefore be determined during calibration before the received signals can be compared directly.

4.3 Calibration Algorithm and Optimisation

The initial calibration algorithm was based on the FFT-based delay estimation method described by Markus Nentwig. This approach uses cross-correlation to estimate the integer-sample delay between two received signals, followed by frequency-domain phase analysis to estimate the remaining fractional delay. Once the signals are aligned, a least-squares complex gain estimate is used to determine their relative amplitude and phase offsets.

Hardware testing identified limitations in the fractional-delay estimation and the use of fixed frequency bounds. The algorithm was therefore refined to improve reliability and support automatic calibration. The final implementation, contained within calibration_optimiser.py, evaluates multiple delay candidates, validates their performance using independent samples and selects the best calibration parameters for continuous processing.

4.3.1 Initial Calibration and Frequency Selection

The calibration signal is generated using signal_gen.py, producing broadband noise between approximately ±10–300 kHz. This is transmitted through the master's TX0 port and distributed to both receivers through the RF switching board in NOISE mode.

Following a 30-second warm-up period, two independent blocks of 100,000 IQ samples are captured, one for calibration estimation and the other for validation. An initial integer-sample delay is determined using FFT-based cross-correlation, as described in the background section.

The original algorithm used a fixed ±300 kHz frequency mask to exclude frequencies outside the calibration signal bandwidth. This was replaced with a dynamic mask that automatically identifies usable frequencies from the received spectra. The power spectra are smoothed, and

frequency bins exceeding the estimated background power by at least 6 dB in both channels are selected. A 20 kHz exclusion around DC and a 10% guard below the Nyquist frequency were used to avoid using potentially unreliable frequency bins.

This allows the calibration algorithm to accommodate different signal bandwidths without manually adjusting the frequency limits.

4.3.2 Calibration Optimisation and Validation

To improve calibration reliability, an optimiser was introduced to evaluate multiple delay candidates and validate their performance before accepting the calibration. Rather than relying solely on the initial cross-correlation peak, the optimiser evaluates five integer-delay candidates, consisting of the initial estimate and its two nearest neighbours in either direction. Each candidate has its fractional delay, amplitude and phase independently estimated.

The raw received signal levels are first checked to ensure that the difference between channels does not exceed 6 dB and that the signal levels remain stable between the estimation and validation captures within 1 dB. Each candidate is then applied to the independent validation samples using the streaming correction algorithm described in Section 4.4.

The following acceptance criteria are used:

- **Correlation magnitude:** ≥ 0.80
- **Normalised root mean square error (NRMSE):** ≤ 0.20
- **Residual integer-sample delay:** 0 samples

Candidates failing these criteria are rejected. The passing candidate with the lowest NRMSE is selected, with correlation used as a secondary criterion. If no candidate passes, the process is repeated using fresh IQ samples, with a maximum of three attempts. If calibration remains unsuccessful, subsequent processing is prevented from starting. This ensures that the estimated calibration parameters are independently validated before being applied during continuous reception and AoA estimation.

4.3.3 Fractional Delay Estimation

The initial fractional delay estimator used phase unwrapping and linear fitting to determine the slope of the relative spectral phase. Testing identified cases where this produced unreliable estimates, so the method was replaced with bounded optimisation.

For each integer-delay candidate, the signals are first aligned and converted into the frequency domain using an FFT. The optimiser then tests different fractional-delay corrections between −0.5 and +0.5 samples, selecting the value that produces the greatest agreement between the two signals across the frequency bins selected by the dynamic mask. This avoids the need for phase unwrapping and linear fitting used in the original algorithm.

Once the fractional delay is determined, it is applied in the frequency domain before estimating the relative amplitude and phase using the existing least-squares complex gain method. The resulting offsets are stored in a Calibration object for correction of the received IQ streams.

4.4 Continuous Calibration Correction

Once the relative delay, amplitude and phase offsets have been determined, they must be applied to the continuously received IQ samples. This is implemented in calibration.py which applies the estimated corrections while preserving sample continuity between processing windows.

This is implemented in calibration.py, which applies the estimated corrections continuously while retaining unprocessed samples and filter information between processing windows. This ensures that the signals remain aligned and that no discontinuities are introduced when moving from one window to the next.

4.4.1 Integer Sample Alignment

The first stage compensates for the integer sample delay between the receivers, which arises primarily because the two SDRs begin acquiring samples independently.

Rather than shifting every processing window individually, the required number of samples is discarded from the appropriate receiver stream once during initialisation. The Calibration class maintains separate buffers for both receivers, allowing any samples left over after processing to be retained and used in subsequent windows. Preventing unnecessary sample loss and maintaining continuous alignment between the two streams.

This ensures that the initial offset is removed without repeatedly discarding samples or interrupting the continuity of the received signals. Once alignment has been established, following windows can be processed continuously.

4.4.2 Fractional Delay Correction

Following integer alignment, a fractional sample timing difference still remains. Since samples can only be shifted by integer positions within an array, the remaining delay is corrected using interpolation. Unlike the finite recordings used in the calibration procedure, where the correction is applied in the frequency domain, continuous reception requires a filter that maintains its state between processing windows.

A 65-tap Kaiser-windowed sinc FIR filter is applied to the slave receiver. The sinc function provides interpolation between discrete sample positions, using a weighted combination of neighbouring samples to estimate the signal at the required fractional position.

The filter coefficients are calculated using:

$$h[n] = \mathrm{sinc}(n - 32 + \delta)\,, w[n]$$

where (δ) is the estimated fractional delay and ($w[n]$) is the Kaiser window. The filter coefficients are normalised so that their sum equals one, preventing any amplitude change to the DC component of the signal.

The 65-tap length was selected as a practical balance between interpolation accuracy, computational cost and processing delay. Longer filters can provide a more accurate approximation of the ideal sinc response, but require additional calculations and introduce greater latency. A 65-tap filter has a centre position of 32 samples, providing a manageable delay while using sufficient neighbouring samples for interpolation. A Kaiser window with a parameter of ($\beta = 8.6$) was selected to smoothly reduce the coefficients towards the ends of the filter, limiting unwanted ripples caused by truncating the infinite sinc function.

The master signal is delayed by a fixed 32 samples, while the slave is delayed by approximately (32-δ) samples. This compensates for the relative fractional delay while allowing both signals to be processed continuously using current and previously received samples. The Kaiser window reduces the effects of truncating the theoretically infinite sinc function to a finite number of coefficients, limiting unwanted ripples in the filter's frequency response.

The filter is implemented using SciPy's lfilter() function. Its internal state is preserved between consecutive processing windows, along with the master's delay history, preventing discontinuities at window boundaries. The first 64 output samples are discarded to remove the initial filter transient, after which the correction operates continuously.

4.4.3 Amplitude and Phase Correction

After time alignment, the remaining relative amplitude and phase differences are corrected using the complex scaling factor (c) determined during calibration.

The slave signal is corrected using:

$$y_{\text{corrected}}[n] = \frac{y_{\text{filtered}}[n]}{c} = y_{\text{filtered}}[n]\frac{e^{-j\phi}}{A}$$

where (A) is the estimated amplitude ratio and (ϕ) is the estimated relative phase offset. The master signal remains the reference, producing two corrected IQ streams for processing.

The calibration parameters remain fixed throughout the measurement period. When transitioning from NOISE to ANTENNA mode, the calibration transmitter is stopped and the RF switching board changes the receiver inputs without restarting acquisition. The existing corrections continue to be applied, with the filter state maintained throughout the transition.

This is particularly important for AoA estimation, as continuously recalculating the relative phase correction from the received antenna signals would remove the spatial phase difference required to estimate the direction of arrival. Any additional phase offset introduced by the antenna signal paths must therefore be characterised separately.

4.5 Calibration Testing Methodology

Testing was conducted to evaluate the reliability of the updated calibration algorithm and the ability of the continuous correction system to maintain receiver alignment over time. This included comparisons of fractional-delay estimation methods, validation of the calibration optimiser, evaluation of dynamic frequency masking, and long-duration stability testing.

Experiments used the two bladeRF receivers with a shared reference clock and a common broadband calibration signal. Earlier tests used a direct RF splitter, while later testing was performed through the RF switching board in NOISE mode, using the hardware configuration described in Section 4.2.

4.5.1 Fractional Delay Estimator Comparison

Initial hardware testing identified cases where the original phase-unwrapping method produced large, inconsistent fractional-delay estimates. To investigate this, test_delay_methods.py was used to compare three methods using identical IQ samples:

- Phase-slope fitting with phase unwrapping.
- Phase-slope fitting without phase unwrapping.
- Bounded coherent-correlation optimisation.

Each method used the same initial integer-sample alignment. The resulting fractional-delay estimates and corrected correlation were compared. Valid estimates were also applied to 6,000 consecutive windows of 10,000 samples, recording correlation, NRMSE, residual delay and phase variation.

4.5.2 Calibration Optimiser Validation

The calibration optimiser was tested through the RF switching board to assess its ability to select suitable calibration parameters. Five neighbouring integer-delay candidates were evaluated using independent estimation and validation samples, as described in Section 4.3. For each candidate, the following were recorded:

- Integer and fractional-delay estimates.

- Corrected correlation and NRMSE.
- Residual integer-sample delay.
- Acceptance or rejection against the calibration criteria.

The selected calibration was then applied to continuous IQ recordings to evaluate its stability, as described in Section 4.5.4.

4.5.3 Dynamic Frequency Masking

The dynamic frequency mask was analysed using broadband calibration signals with bandwidths of ±250, ±300 and ±350 kHz, all other calibration algorithms were unchanged.

For each bandwidth, the selected frequency bins, fractional-delay estimates, correlation and NRMSE was calculated over 60 seconds of continuous reception. This assessed whether the dynamic mask could adapt to different signal bandwidths without reducing calibration performance.

4.5.4 Continuous Correction and Stability Testing

The continuous correction was initially tested by applying calibration parameters to newly acquired IQ samples, verifying that integer-delay, fractional-delay, amplitude and phase corrections remained effective across consecutive processing windows.

Long-term stability was then evaluated using the following experiments:

- **Direct RF splitter:** Tests ranging from several seconds to 20 minutes, with a 10-second warm-up introduced during development.
- **RF switching board:** 60-second and 180-second tests using the calibration optimiser and a 30-second warm-up.

For each experiment, calibration parameters remained fixed throughout reception. IQ data was processed in 10,000-sample windows (10 ms at 1 MS/s), recording:

- Residual integer and fractional-sample delay.
- Corrected amplitude ratio and relative phase.
- Correlation, NRMSE and residual SNR.
- Processing time and receiver queue occupancy.

These measurements were used to assess calibration drift, alignment stability and continuous processing performance. Residual fractional-delay estimates were treated as diagnostics rather than independently verified timing errors.

4.6 RF Switching Board Integration and Validation

The supplied RF switching board was unvalidated, so its operation was tested before integration into the dual-SDR calibration system. Testing began with an individual SDR, followed by dual-receiver calibration and switching repeatability experiments.

4.6.1 Initial RF Path Validation

Initial testing used a single bladeRF to transmit and receive a known signal through the board. The following settings were used:

- RF centre frequency: 100 MHz.
- Transmitted tone offset: 125 kHz.
- Sampling rate: 1 MS/s.

- TX gain: 0 dB, digital amplitude: 0.02.
- RX gain: 10 dB.

The board was controlled using CaliBoard.py, with the indicator LED used to confirm the selected operating mode.

In NOISE mode, the tone was injected through J2 and measured at the SDR1 and SDR2 outputs individually. Measurements were taken with transmission enabled and disabled to establish whether the calibration signal reached each output. An additional test disconnected the transmitter from J2 to distinguish the received signal from internal SDR leakage.

In ANTENNA mode, the tone was injected through ANT1 and ANT2 to verify their corresponding SDR outputs. Both master and slave bladeRF devices were used during the individual path tests. The board was also cycled between NOISE and ANTENNA modes, with the transmitter disabled during switching. Received tone levels were compared before and after switching to assess signal-path continuity and repeatability.

4.6.2 Dual SDR Integration

Following individual path testing, the board was integrated into the existing dual-SDR calibration setup, replacing the external RF splitter.

The master transmitted broadband calibration noise through J2, while SDR1 and SDR2 supplied the master and slave receiver inputs respectively. Both receivers operated using the shared reference clock.

The original calibration pipeline was used to estimate and correct the relative delay, amplitude and phase offsets. Correlation, NRMSE and residual timing were measured to verify simultaneous calibration through the board and provide a comparison with the previous splitter configuration.

4.6.3 RF Switching and Stability Testing

Further experiments investigated whether calibration remained effective when transitioning between NOISE and ANTENNA modes. An initial control test stopped and restarted the calibration transmitter without changing the RF switch position, allowing transmitter effects to be investigated separately from switching effects.

For further switching experiments, slave TX0 supplied a common broadband signal through an RF splitter to ANT1 and ANT2. Calibration was performed in NOISE mode, followed by a five-second baseline measurement. The master transmitter was then stopped, and the board was switched to ANTENNA mode.

Two experiments were performed:

- **Switching repeatability:** Ten return-switching cycles, with five seconds of corrected IQ recorded after each return to ANTENNA mode.
- **Phase stability:** A 180s ANTENNA mode recording without further switching, to distinguish gradual phase drift from changes associated with switching.

The original calibration parameters remained fixed throughout both experiments. Residual phase, amplitude ratio, correlation, NRMSE and integer-sample delay were recorded. The final operating procedure also disables RF transmission before switching modes to protect the board's reflective RF switches.

4.7 Phase-Offset Investigation Methodology

Following the RF switching validation, further experiments were conducted to investigate observed random variations in the residual phase offset. These offsets would appear between NOISE and ANTENNA modes across independent trials. Each experiment used a new initial calibration, which remained fixed during the following measurements.

4.7.1 Switching and Startup Repeatability

Initial testing compared manual and automatic RF switching across five independent runs of each method. Each run involved calibration in NOISE mode, followed by measurements in ANTENNA mode and five return-switching cycles.

The residual phase offset, phase variation, amplitude ratio, correlation and remaining integer-sample delay were recorded. Additional tests examined the effect of stopping the master transmitter without switching RF paths, to distinguish transmitter-related changes from switching effects.

4.7.2 Receiver and RF Source Investigation

To investigate the source of the phase variation, two alternative hardware configurations were tested:

- **Single-SDR reception:** Both RF board outputs were connected to RX0 and RX1 of the master bladeRF, using simultaneous dual-channel acquisition. Two independent runs were conducted to compare phase-offset repeatability against the original two-SDR configuration.
- **External noise source:** A BG7TBL broadband noise generator replaced the slave transmitter as the ANTENNA-mode signal source. Its output passed through 32 dB of attenuation and an RF splitter before reaching ANT1 and ANT2. Five independent calibration and measurement runs were completed.

In the external-source experiments, RF transmission was disabled during switching, while receiver acquisition and the existing calibration correction continued uninterrupted. The received signals were also analysed across frequency to check for any remaining timing errors and determine whether the phase difference was consistent across the signal bandwidth.

4.8 AoA Estimation and Simulation

4.8.1 AoA Estimation Algorithm

A one-dimensional angle-of-arrival (AoA) estimator was developed using the relative phase between two calibrated receiver signals. Phase comparison was selected for its compatibility with the existing two-channel configuration and relatively low computational complexity.

The relative phase between the master and slave IQ signals is calculated using:

$$\Delta\phi = \arg\left(\sum_{n=0}^{N-1} x^*[n]y[n]\right)$$

where $(x[n])$ and $(y[n])$ are the calibrated master and slave signals, respectively. Ideally, with the relative time delay, amplitude and hardware phase offsets compensated for, the remaining phase difference represents the signal's angle of arrival. However, any residual phase offset not accounted for during calibration will contribute to the measured phase difference and introduce an error in the estimated arrival angle.

The phase difference is converted into an arrival angle relative to array broadside (Figure 4) using the relationship established in Section 3.2. The implementation defines positive arrival angles as producing negative slave-relative phase differences.

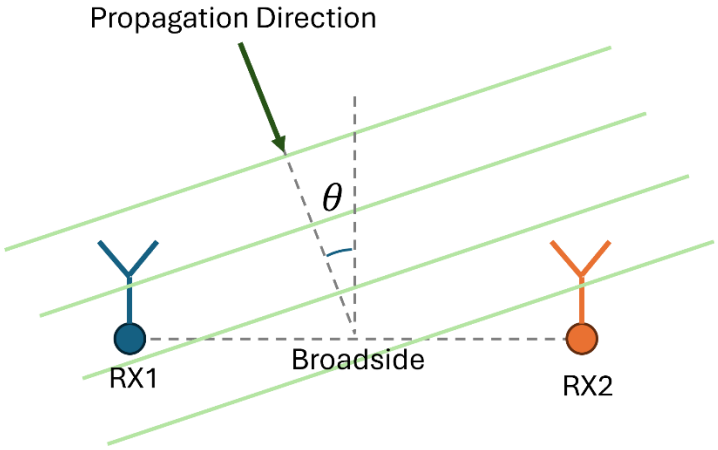


Figure 4: Illustration of the angle-of-arrival (AoA) geometry for two spatially separated receiver antennas, showing the incoming plane wave and its angle relative to broadside.

The estimator also calculates the normalised correlation magnitude:

$$\rho = \frac{|\sum_n x^*[n]y[n]|}{\sqrt{\sum_n |x[n]|^2 \sum_n |y[n]|^2}}$$

This normalises the complex cross-correlation by the energy of each signal, producing a value between zero and one. Values approaching one indicate strong agreement between the received signals, independently of constant amplitude and phase differences. During continuous AoA estimation, this metric is used to assess the reliability of each processing window. Windows with a correlation below 0.80 are rejected, preventing potentially unreliable phase measurements from being used to produce AoA estimates.

4.8.2 AoA Simulation Methodology

The AoA estimator was validated using synthetic IQ signals with known arrival angles. Each signal pair was generated with the expected geometric phase difference, allowing the estimated angle to be compared against the true value.

The simulation used the following configuration:

- RF frequency: 915 MHz.
- Antenna spacing: 0.164 m (approximately half a wavelength).
- Two-element linear antenna array.
- Independent complex additive white Gaussian noise (AWGN).
- 120 Monte Carlo trials per condition.

Initial testing swept arrival angles from −75° to +75° under noise-free conditions, followed by a 10 dB SNR test using 10,000 samples. Further Monte Carlo testing evaluated angular RMSE at a fixed arrival angle of 30°, varying receiver SNR and sample count.

Additional simulations investigated the effect of residual hardware phase offsets identified during RF switching experiments. The measured phase was modelled as:

$$\Delta\phi_{\text{measured}} = -\frac{2\pi d}{\lambda}\sin\theta + \phi_{\text{hardware}}$$

where (ϕ_{hardware}) represents an unknown phase offset selected once per simulated session. Uniform phase-offset ranges of ±7.5° and ±15° were tested to assess their effect on AoA accuracy. A secondary reference calibration was also investigated in simulation. A signal arriving from a known angle of 0° was used to estimate the hardware phase offset, which was then subtracted from subsequent measurements:

$$\Delta\phi_{\text{corrected}} = \Delta\phi_{\text{target}} - \Delta\phi_{\text{reference}}$$

The reference calibration was evaluated using different sample counts and additional phase drift of up to ±2° to assess its sensitivity to noise and changing hardware conditions. These tests evaluated the estimator's accuracy under ideal calibration, unknown hardware phase offsets and simulated reference correction. The secondary reference calibration was not implemented or validated on the physical system.

4.9 Full Software Pipeline

The final software system integrates the hardware configuration, calibration, continuous correction and AoA estimation stages into a single processing pipeline. The full workflow is shown in Figure 5. This structure allows the system to perform initial calibration in NOISE mode, maintain the resulting correction during continuous reception, and subsequently estimate AoA from the calibrated antenna signals.

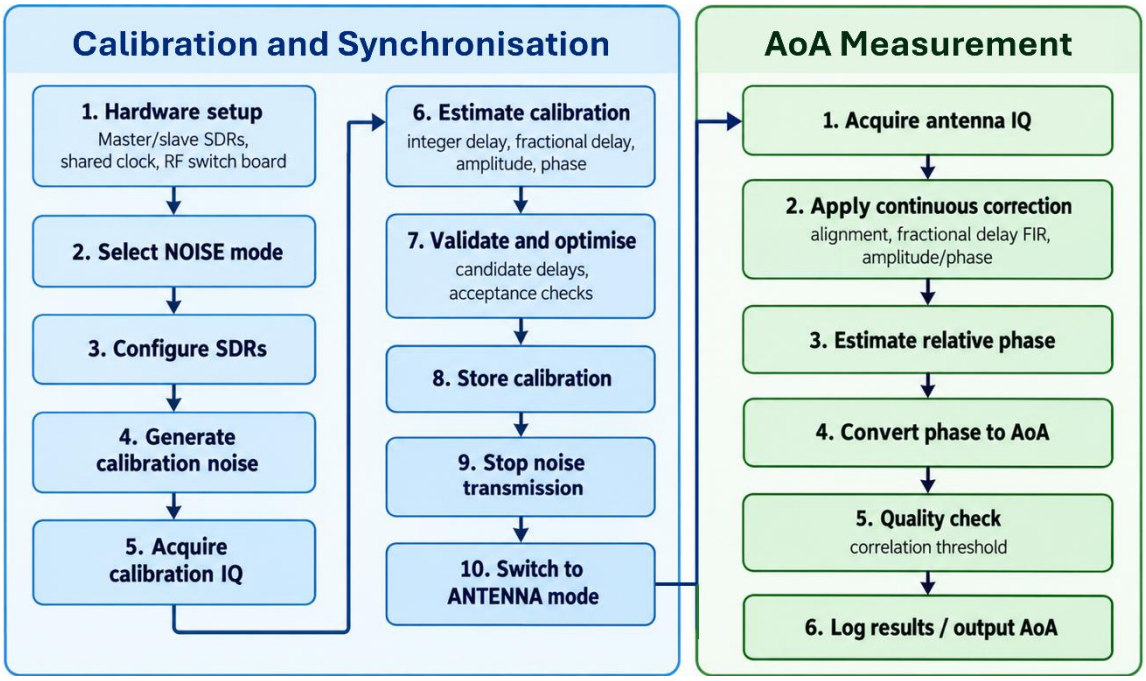


Figure 5: Full software pipeline from hardware setup to AoA.

5 Results and Evaluation

5.1 Initial Calibration Algorithm Validation

Requirements Addressed – R4.1, R4.2, R4.3, R4.5

Initial testing evaluated the calibration algorithm's ability to recover known time-delay, amplitude and phase offsets, and examined its performance under varying noise conditions and during hardware operation.

The main test conditions were:

- **Ideal simulation:** 1,000 trials using 100,000 samples with randomly generated offsets and no added noise.
- **Monte Carlo simulation:** 100 trials per condition, varying SNR and sample count.
- **Hardware validation:** Single bladeRF TX/RX loopback, followed by independent startup and continuous acquisition tests.

The key results from the progress report are summarised in Table 2.

Table 2: Core tests carried out in the initial validation, outlined in more depth in the progress report.

Test	Key results
Ideal simulation	Near-zero estimation error under noise-free conditions.
Monte Carlo, 30 dB SNR	**NRMSE: 0.0317** at 100,000 samples; minimal improvement beyond 100,000 samples.
Hardware loopback, 3 trials	**Correlation:** 0.9114–0.9138; **NRMSE:** 0.0299–0.0336; **Residual SNR:** 29.47–30.50 dB; **Noise-floor SNR:** 30.86 dB.
50 independent hardware starts	**43/50** trials retained; **Mean residual SNR:** 29.84 dB; phase varied between starts.
Continuous 60-second acquisition	**Phase estimate standard dev:** 0.00047°; **Total-delay estimate standard dev:** 0.0004 samples; **Mean residual SNR:** 30.39 dB.

The simulation results demonstrated that the algorithm could accurately recover known channel offsets, with estimation error decreasing as SNR and sample count increased. At 30 dB SNR, performance approached the measured hardware noise-floor SNR of 30.86 dB, while increasing the sample count beyond 100,000 provided limited additional improvement.

Hardware testing demonstrated consistent calibration performance during continuous acquisition, with residual SNR approaching the measured noise floor. However, only 43 of the 50 independent startup trials were retained, and the estimated phase varied between starts. In contrast, the uninterrupted 60-second experiment produced a phase-estimate standard deviation of approximately 0.027° and total-delay standard deviation of 0.0004 samples. This established that the estimated offsets were highly stable within an individual acquisition, but were not necessarily repeatable between independent starts.

5.2 Dual-SDR Synchronisation and Continuous Correction

Requirements Addressed – R1, R3.2, R4.5

Testing was conducted to verify that two bladeRF receivers could be synchronised using a shared reference clock, and that their initial calibration could be applied continuously to subsequently received IQ samples.

The main test conditions were:

- **Hardware:** Two bladeRF SDRs connected to a common 38.4 MHz clock reference, receiving a shared broadband signal through an RF splitter.
- **RF centre frequency:** 100 MHz.
- **Sample rate:** 1 MS/s.
- **Processing window:** 10,000 samples.
- **Calibration:** 100,000 samples, with correction parameters held constant during subsequent reception.

- **Stability testing:** Up to 20 minutes, with a 10-second warm-up before calibration (Figure 6).

Initial streaming validation was performed over 500 consecutive processing windows (5 seconds) to assess the effectiveness and computational performance of the calibration corrections. The results are summarised in Table 3. The high corrected correlation and low NRMSE indicate effective signal alignment, while the average processing time of 1.7 ms per window was comfortably below the 10 ms acquisition interval.

To investigate longer-term stability, a separate 20-minute continuous acquisition test was performed, with a 10-second warm-up before calibration. The resulting calibration diagnostics are shown in Figure 6.

Table 3: Initial validation test results across 500 processing windows.

Metric (Mean)	**Result**
Correlation	0.998799
NRMSE	0.04941
Residual Phase	-0.1509°
Processing Time	1.7ms/window

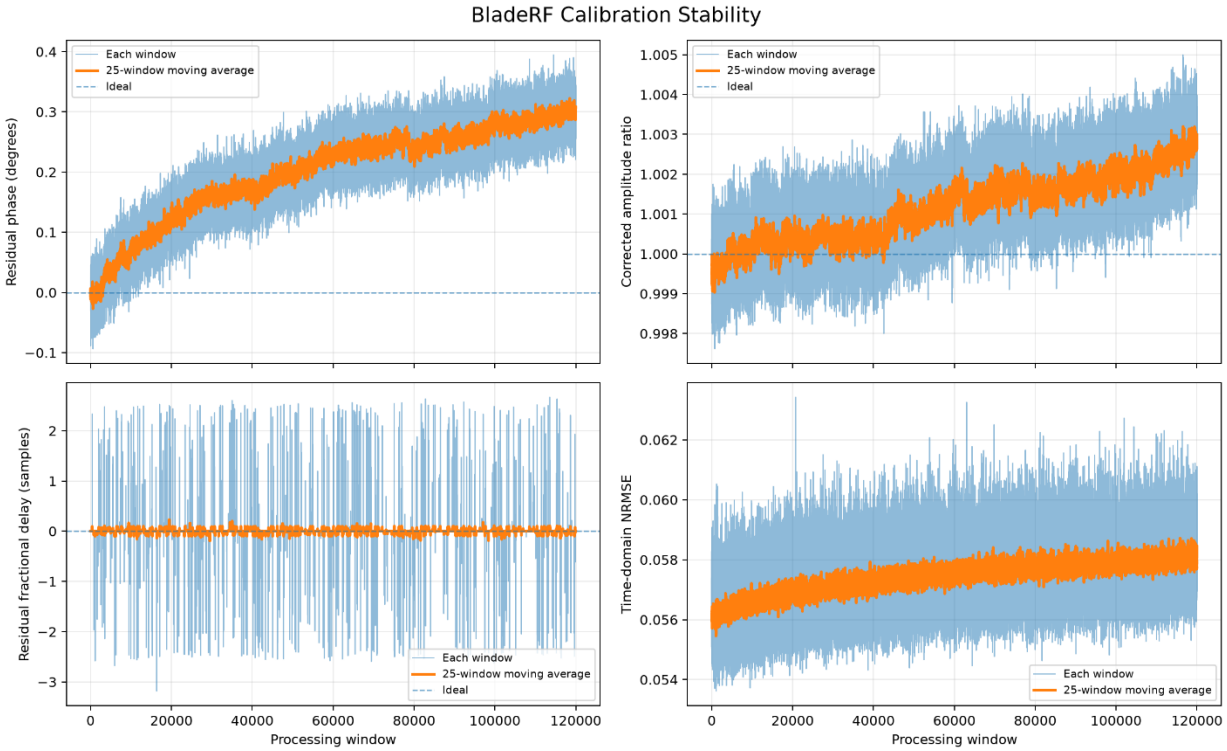


Figure 6: Residual calibration diagnostics over a 20 minute continuous run.

Residual Phase (Top Left) - Increased gradually from ~ 0° to +0.30°

Amplitude Ratio (Top Right) - Increased from ~ 1.000 to 1.003

Residual Fractional Delay (Bottom Left) – Centred around 0, with occasional outliers (Section 4.5.1)

Time-domain NRMSE (Bottom Right) – Increased slightly from ~ 0.056 to 0.0585

These results demonstrate that the two BladeRF receivers could be synchronised and their relative offsets compensated for during continuous acquisition. However, small residual phase

and amplitude changes accumulated over time despite the calibration remaining unchanged. These changes may be associated with thermal drift, although further controlled testing would be required to establish the cause.

5.3 Initial RF Switching Board Validation

Requirements Addressed – R3.1, R3.2

The RF switching board was supplied without prior validation. Its NOISE and ANTENNA signal paths were therefore tested individually before integration into the dual-SDR calibration system. Initial tests used a single bladeRF for transmission and reception, with both master and slave devices tested separately. The board was controlled using CaliBoard.py, with the following configuration:

- **NOISE mode:** TX0 → J2 → SDR1/SDR2 → RX0 (green LED on).
- **ANTENNA mode:** TX0 → ANT1/ANT2 → SDR1/SDR2 → RX0 (green LED off).
- **Test signal:** 125 kHz tone offset from a 100 MHz centre frequency.
- **Settings:** 1 MS/s sampling rate, 0 dB TX gain and 10 dB RX gain.

Received tone levels were measured with transmission enabled and disabled to verify signal-path continuity. Representative results are shown in Table 4.

Table 4: Comparison of signal dBFS between the noise and antenna paths for inputs 1 and 2.

Board Mode	RF Path	Received Signal (dBFS)
NOISE	J2 -> SDR1	-83.15
NOISE	J2 -> SDR2	-84.45
ANTENNA	ANT1 -> SDR1	-65.27
ANTENNA	ANT2 -> SDR2	-64.96

In NOISE mode, disabling the transmitter or disconnecting the J2 injection path reduced the received tone to approximately −102 dBFS. The ANTENNA-mode paths produced tone levels approximately 36–38 dB above their corresponding transmitter-off measurements.

The received tone was approximately 18–20 dB lower through the NOISE-mode path than through the ANTENNA-mode path. However, these measurements used different external RF connections, as the primary objective was to verify individual signal-path continuity rather than characterise the full path. Therefore the cause of the observed difference was not investigated further.

Following individual signal-path validation, the RF switching board replaced the external splitter in the dual-SDR calibration system. The master SDR transmitted broadband calibration noise through J2, while SDR1 and SDR2 supplied the master and slave receivers, respectively. Both receivers shared the master's 38.4 MHz clock reference.

An initial NOISE-mode calibration test achieved a mean corrected correlation of 0.9914, NRMSE of 0.1315 and residual phase standard deviation of 0.0398°, with no residual integer-delay errors across 20 processing windows (0.2 seconds). These results confirmed that the RF switching board could distribute a common calibration signal to both receivers, with the corrected signals exhibiting high correlation and no residual integer-delay errors during the initial test. Combined with the individual signal-path measurements, this established the board's suitability for integration into the dual-SDR calibration system.

5.4 Calibration Algorithm Improvements and Validation

Requirements Addressed – R2.3, R3.1, R3.2, R4.5, R4.6

The updated calibration algorithms were evaluated using the dual-BladeRF system and the RF switching board in NOISE mode. The hardware configuration for the following experiments is similar to that seen in Figure 3, however as this was done entirely in NOISE mode there was nothing connected to the ANT1/2 inputs on the switch board.

5.4.1 Fractional-Delay Estimator Comparison

Three fractional delay methods were tested and compared to investigate the large outliers seen in Figure 5 (bottom left graph). The original phase-unwrapping approach is compared against bounded correlation optimisation as an alternative to reduce these outliers.

The following tests used identical IQ samples with the same integer delay alignment applied. The methods that produced a valid estimate were then applied to 6,000 processing windows (60s) of 10,000 samples.

Initial estimation results are shown in Table 5.

Table 5: Comparison of different fractional delay estimation methods.

Estimation Method	Fractional Delay (Samples)	Corrected Correlation
Phase fitting (unwrapping)	+4.9807	Not evaluated
Phase fitting (no unwrapping)	+0.3187	0.9953
Bounded Correlation Optimisation	+0.3182	0.9953

Both phase fitting without unwrapping and bounded correlation optimisation produced nearly identical fractional-delay estimates and correction performance, indicating that incorrect phase unwrapping was responsible for the large estimation error observed with the original method. During the 60-second tests, both alternatives achieved a mean corrected correlation of 0.9919 and mean NRMSE of approximately 0.1277, with residual fractional-delay estimates remaining within ±0.5 samples. Bounded correlation optimisation was selected for the final calibration algorithm because it avoids reliance on phase unwrapping while maintaining equivalent correction performance.

Figure 7 below shows the results from a later test with the bounded correlation optimiser implemented. The spread in fractional delays across this 60 second test are all bounded to within +/-0.3, a much more reasonable range than the previous +/-3 samples.

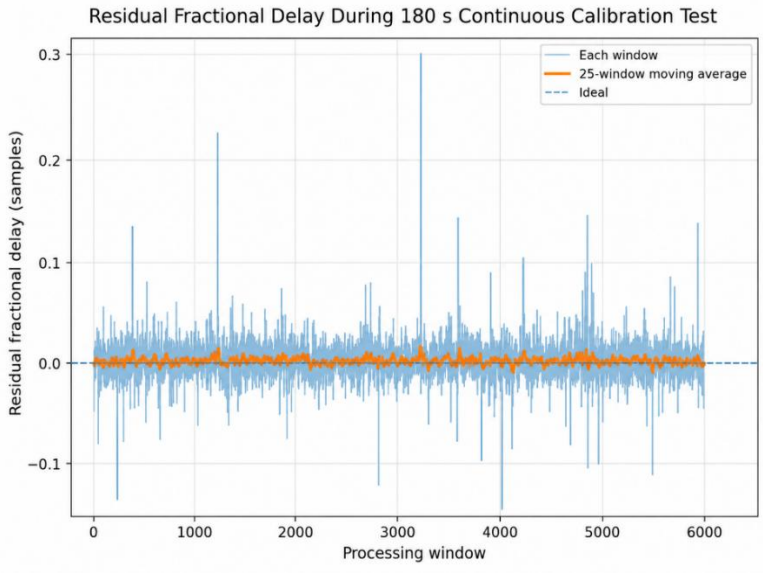


Figure 7: Fractional Delay spread over a 180 second (18,000 window) calibration test.

5.4.2 Calibration Optimiser Validation

An earlier 60-second calibration test showed a persistent residual integer delay of +1 sample, with a mean corrected correlation of 0.8244 and NRMSE of 0.6880. Further diagnostic testing evaluated additional integer shifts from −3 to +3 samples. As shown in Figure 8, the original alignment produced the highest correlation of 0.9887, while applying an additional +1-sample shift reduced correlation to 0.5457. These results demonstrated the sensitivity of calibration performance to integer-sample alignment and motivated the introduction of candidate-delay validation.

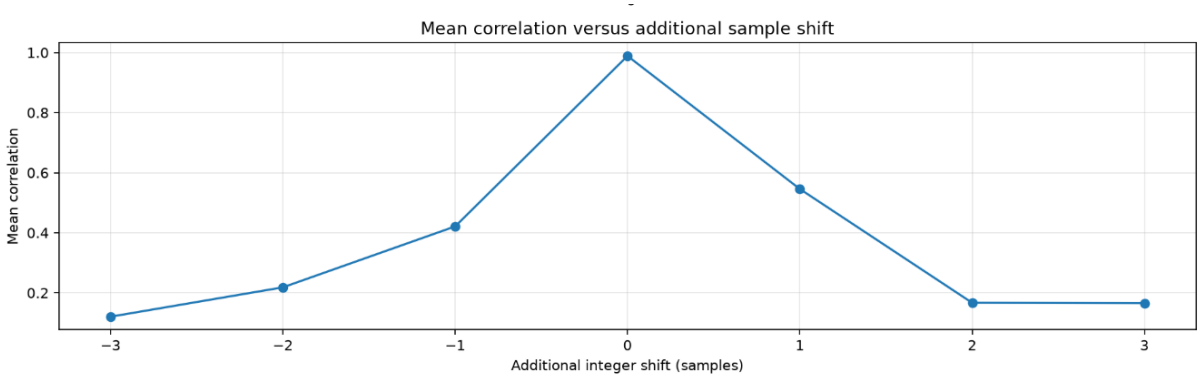


Figure 8: Effect on correlation of an integer delay offset.

The calibration optimiser was therefore implemented to evaluate four neighbouring integer-delay candidates and validate the resulting corrections against independent IQ samples. Following calibration selection, the saved correction was applied to newly acquired IQ samples over 60-second and 180-second experiments to evaluate continuous performance. The results are summarised in Table 6.

Table 6: Comparison between a 60 and 180 second test using core metrics.

Metric	60 Second test	180 Second test
Num Windows	6,000	18,000
Mean corrected correlation	0.9925	0.9906
Mean NRMSE	0.1228	0.1376
Residual phase std dev	0.0539°	0.0819°
Non-zero residual integer delay	0	0

Both continuous tests achieved mean corrected correlations above 0.99, with no detected residual integer-delay errors across the 6,000 and 18,000 processing windows. The 180-second experiment also exhibited gradual residual phase drift of approximately +0.2°.

During calibration optimisation, the acquisition threads continued receiving IQ samples, producing a temporary backlog in the receiver queues. Peak occupancy reached 48 of the available 256 blocks, before returning to near zero once continuous processing began. Mean processing times were approximately 1.53 ms and 1.63 ms per 10 ms window for the 60-second and 180-second tests, respectively.

These results demonstrate that the optimiser could select suitable calibration parameters and maintain effective sample alignment during acquisition. Processing times were generally below the window duration, allowing the temporary acquisition backlog to clear without sustained accumulation.

5.4.3 Dynamic Frequency Masking

The dynamic frequency masking algorithm was tested to evaluate its ability to automatically identify the usable frequency range for calibration signals of different bandwidths. Tests used broadband noise signals with bandwidths of ±250, ±300 and ±350 kHz, with each calibration applied to 60 seconds of continuous reception (6,000 processing windows).

Table 7: Dynamic frequency masking performance against different bandwidths (BW).

Transmitted BW	Detected BW	Mean Corrected Corr	Mean NRMSE
+/-250 kHz	+/-252.4 kHz	0.9917	0.1293
+/-300 kHz	+/-302.4 kHz	0.9923	0.1240
+/-350 kHz	+/-352.4 kHz	0.9920	0.1264

The dynamic mask identified all three signal bandwidths within 2.4 kHz of their nominal limits, maintaining mean corrected correlations above 0.99 and NRMSE below 0.13 across the experiments (Table 7). Comparison against the original fixed ±300 kHz mask using identical IQ samples showed negligible differences in calibration performance, with mean NRMSE differing by less than 0.000061 and fractional-delay estimates differing by no more than approximately 0.0008 samples. These results demonstrate that dynamic masking can adapt to different calibration-signal bandwidths without degrading correction performance under the tested conditions. Its primary advantage is therefore adaptability rather than improved calibration accuracy.

5.5 RF Switching and Stability Validation

Requirements Addressed – R3.2, R4.5, R4.6

Following successful RF board integration and calibration algorithm validation, further testing investigated whether the saved calibration remained effective when switching between NOISE and ANTENNA modes. Initial experiments used the slave bladeRF transmitter to supply a common ANTENNA-mode signal through an RF splitter, rather than the external BG7TBL noise source used in later experiments.

The hardware configuration was:

- **Master TX0 → J2:** Common NOISE-mode calibration signal.
- **Slave TX0 → RF splitter → ANT1 and ANT2:** Common ANTENNA-mode test signal.
- **SDR1 and SDR2 → master and slave RX0:** Continuous dual-receiver acquisition.
- **Master CLKOUT → slave CLKIN:** Shared clock reference.

An initial control test found that restarting the master transmitter while receiver acquisition remained active reduced correlation from 0.9915 to 0.0109, without switching the board. Subsequent tests therefore started both transmitters before acquisition, stopped the master transmitter after calibration, and kept the slave transmitter active throughout switching.

Ten switching cycles were performed without recalibration, with the original NOISE-mode calibration applied continuously. A separate 180-second experiment was conducted with the board held in ANTENNA mode to provide a comparison without repeated switching.

Table 8: Results from key metrics for two tests, one covering 10 switching cycles and one focused on stability without switching.

Metric	10 Switching Cycles	Stability Test (180s)
Switching cycles	10	0
Corrected correlation	0.99893	0.99901

Residual phase change	+0.3096°	+0.2022°

Both tests maintained high corrected correlation and zero residual integer-delay errors. Gradual residual phase changes were observed in both experiments, including the test without switching. A caveat to this approach is it involves switching while the slave transmitter remains active. Due to the boards reflective RF switches a safer approach was taken for later experiments, ensuring all transmission is disabled when switching.

5.6 Calibration Transfer and Phase Repeatability

Requirements Addressed – R3.2, R4.5, R4.6

Following RF switching validation, further experiments investigated the repeatability of the residual phase offset when transferring calibration from NOISE to ANTENNA mode. Although corrected signal correlation remained high, the phase offset differed between independent system starts.

5.6.1 Independent Startup Repeatability

Ten independent dual-SDR experiments were performed, comprising five manually switched and five automatically switched runs. Each used a fresh NOISE-mode calibration, followed by ANTENNA-mode measurements and five return-switching cycles. The slave transmitter supplied a common broadband signal to ANT1 and ANT2 through an RF splitter, as described in Section 5.5.

Figure 9 shows the noise to antenna residual phase offset measured in each run. The offsets ranged from −11.015° to +4.069°, giving a total spread of 15.084°. The run-to-run standard deviation was 5.418° for manual switching and 2.449° for automatic switching. However the sample size is too small to definitively determine whether automatic is better than manual. Although a phase difference is expected between the NOISE and ANTENNA signal paths, a fixed path-dependent offset should remain relatively consistent under unchanged operating conditions. This observed variation between independent starts therefore suggests an additional session-dependent contribution to the residual phase offset.

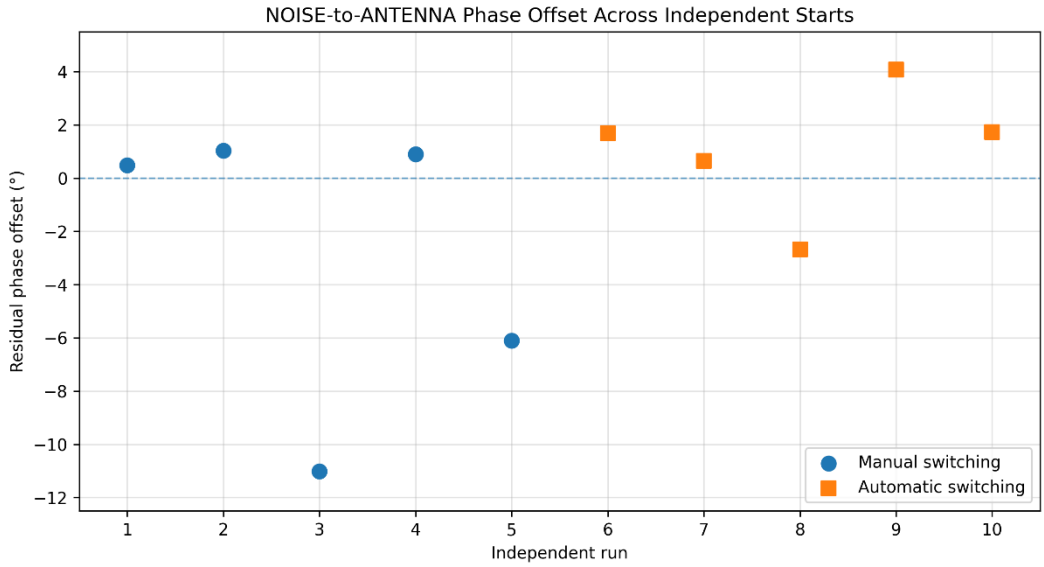


Figure 9: Distribution of residual phase error after switching from NOISE to ANTENNA.

Despite the variation between starts, the initial experiments maintained a mean corrected ANTENNA correlation of 0.99891, with no residual integer-delay errors across 35,000 processing windows, including NOISE mode measurements. The corrected ANTENNA

amplitude ratio also varied between runs, ranging from approximately 0.775 to 1.254. The phase offset was comparatively stable within individual runs, with the larger differences appearing between independent starts rather than during repeated switching.

5.6.2 Receiver and RF Source Investigation

Several experiments were performed to investigate the source of the observed NOISE-to-ANTENNA phase variation.

Test 1: Master Transmitter Shutdown

Purpose: Determine whether disabling master TX0 introduces a relative phase change independent of RF switching.

Method: The RF board remained in ANTENNA mode, with slave TX0 supplying the common signal, while master TX0 was disabled without changing the RF path.

Results: Three independent tests showed phase changes of approximately +0.2° to +0.3°, with correlation remaining near 0.984. Transmitter shutdown therefore produced a measurable phase change, but one substantially smaller than the 15.084° variation observed between independent starts.

Test 2: Transmitter Startup After RF Switching

Purpose: Determine whether calibration could be preserved when both RF sources were disabled during switching, avoiding transmission through the board's reflective RF switches.

Method: Following NOISE-mode calibration, master TX0 was disabled, the board was switched to ANTENNA mode, and slave TX0 was started to provide the common reference signal.

Results: Corrected correlation fell to 0.0097 and 0.0129 in two attempts, with residual integer-delay checks failing across all 500 measurement windows in each attempt. Subsequent recalibration attempts also failed to establish reliable alignment. The revised procedure therefore did not preserve calibration when slave TX0 was started after receiver acquisition, preventing a valid phase-offset comparison.

Test 3: Single-SDR Dual-Channel Reception

Purpose: Investigate whether using two independent SDR receivers contributed to the variation between starts.

Method: Both RF board outputs were connected to RX0 and RX1 of the same bladeRF, using simultaneous dual-channel acquisition instead of two separate SDRs.

Results: Two independent starts produced NOISE-to-ANTENNA phase offsets of +1.2529° and +1.4300°, differing by only 0.1771°, with correlation remaining approximately 0.999. A subsequent 300-second measurement recorded a phase standard deviation of 0.0207°. This configuration demonstrated substantially more repeatable phase offsets in the two trials performed, although the limited sample size did not establish the underlying cause.

Test 4: Independent External Noise Source

Purpose: Determine whether the slave SDR transmitter contributed to the inconsistent phase offset and validate calibration transfer using the revised source-off switching procedure.

Method: Slave TX0 was replaced with a BG7TBL broadband noise generator connected through 32 dB of attenuation and an RF splitter to ANT1 and ANT2 (Figure 3). Both RF sources were disabled during switching, and five independent calibration-transfer experiments were performed.

Results: Residual phase offsets ranged from +0.786° to +9.421°, giving an 8.635° range. Within-run phase standard deviations were 0.067–0.193°, while corrected ANTENNA correlation ranged from 0.9242 to 0.9912. No residual integer-delay errors occurred across 30,000 measurement windows. The variation therefore persisted without the slave transmitter, confirming that it was not necessary for the observed phase-offset inconsistency.

Test 5: Frequency-Domain and Residual-Delay Analysis

Purpose: Investigate whether residual fractional delay or frequency-dependent phase differences could explain the remaining calibration-transfer errors and reduced correlation.

Method: Two additional BG7TBL experiments measured residual delay using averaged cross-spectra, alongside frequency-domain phase and coherence measurements.

Results: Estimated residual delays were −0.000262 and +0.000509 samples, with median spectral coherence of 0.8619 and 0.8958, respectively. The approximately flat phase response suggested that substantial residual baseband delay was unlikely to explain the observed variation. Raw slave-to-master amplitude ratios also differed between NOISE-mode calibrations (0.861 and 1.029), compared with 0.996 and 0.997 in ANTENNA mode. These measurements identified additional differences between calibration sessions but did not isolate the source of the phase offset.

Evaluation:

Overall, these investigations demonstrated that NOISE-mode calibration could maintain effective sample alignment and relatively stable residual phase during individual acquisitions, but did not consistently compensate for the phase offset introduced when transferring to ANTENNA mode. The variation persisted when an independent external RF source was used, while residual-delay measurements indicated that incomplete time alignment was unlikely to be the primary cause. The single-SDR dual-channel configuration exhibited improved repeatability in the limited trials performed, although the underlying cause could not be isolated. Consequently, the remaining startup-dependent phase offset represents a significant limitation for absolute AoA estimation, as any uncompensated hardware phase difference would be interpreted as part of the signal's arrival phase.

5.7 AoA Algorithm Validation

Requirements Addressed – R5, R6

The AoA estimator was validated using synthetic IQ signals with known arrival angles. The results evaluate its accuracy under ideal calibration, sensitivity to residual hardware phase offsets and performance with a simulated secondary reference correction.

5.7.1 Initial Algorithm Validation

Figure 10 compares the estimated and true arrival angles over the range −75° to +75°, alongside angular RMSE for varying receiver SNR and sample count.

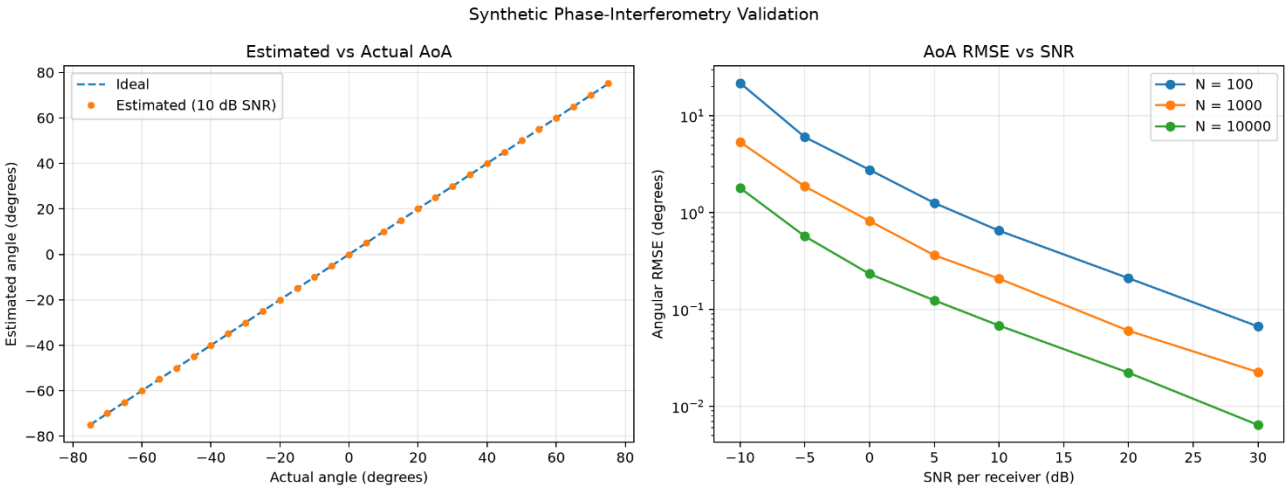


Figure 10: (Left) Estimated angle via AoA vs actual simulated angle. (Right) Angular RMSE over different SNR levels.

The estimated angles closely followed the ideal relationship at 10 dB SNR using 10,000 samples. Monte Carlo testing also showed decreasing angular RMSE with increasing SNR and sample count. At 10 dB SNR, increasing the sample count from 100 to 10,000 reduced RMSE from 0.754° to 0.070°. At 10,000 samples, increasing SNR from 0 to 20 dB reduced RMSE from 0.250° to 0.023°. These results validate the estimator's ability to recover known arrival angles under ideal calibration conditions, with angular accuracy improving as the available signal quality and sample count increase.

5.7.2 Sensitivity to Residual Hardware Phase Offset

The calibration-transfer experiments in Section 5.6 identified a 15.084° total variation in residual phase offset between independent system starts. Simulations therefore investigated the effect of an unknown phase offset using uniform ±7.5° and ±15° distributions on estimator accuracy (Figure 10). The ±7.5° range approximately represents the spread observed experimentally, while ±15° provides a more conservative test condition.

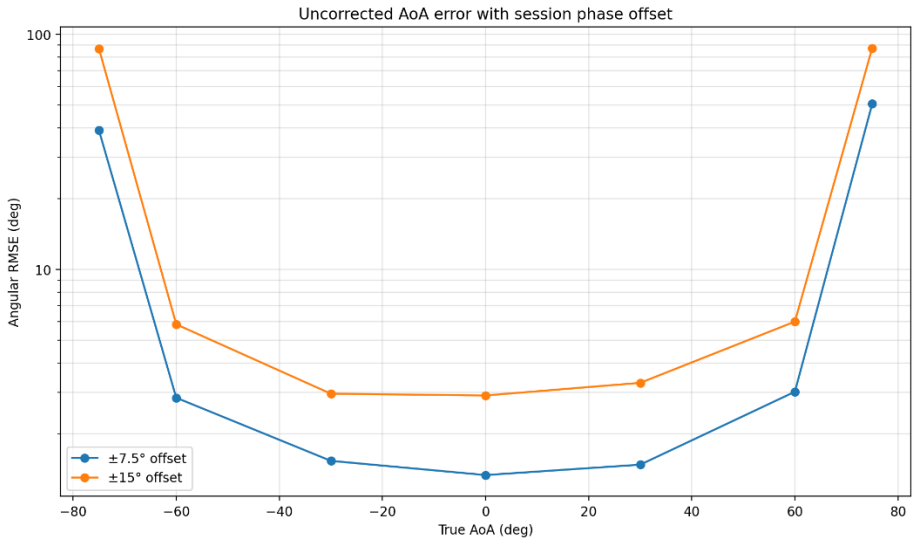


Figure 11: RMSE vs True AoA angle with a uniform phase offset applied.

At 30° AoA, 10 dB SNR and 5,000 samples, the ±7.5° phase-offset model increased angular RMSE from approximately 0.1° under ideal calibration to 1.477°. The effect became more pronounced near the edges, where phase wrapping produced errors approaching 40° RMSE at

±75°. These results demonstrate that an uncompensated hardware phase offset can substantially reduce AoA accuracy, even when the underlying estimator performs well under ideal calibration.

A secondary calibration using a simulated known-angle reference was also investigated as a potential means of compensating for the unknown phase offset. At 30° AoA, 10 dB SNR and 5,000 samples, this reduced angular RMSE from 1.477° to 0.147°. However, this approach was validated only in simulation, and a practical method of obtaining a suitable phase reference was not established.

5.8 Integrated System Validation

Requirements Addressed – R1, R3.2, R4.5, R5

Following individual algorithm validation, the complete software pipeline was tested using the dual-SDR system and RF switching board. The system performed NOISE-mode calibration, disabled transmission before switching to ANTENNA mode, and continuously applied the saved calibration to incoming IQ samples before AoA estimation.

For this experiment, the BG7TBL external noise source was connected as in Figure 3 testing setup, providing a common signal to both receivers. The following configuration was used:

- **RF centre frequency:** 100 MHz.
- **Sampling rate:** 1 MS/s.
- **AoA calculation:** 10,000 samples per processing window.
- **Configured antenna spacing:** 1.499 m.
- **Secondary phase correction:** Disabled.

The final experiment processed 18,000 consecutive windows, corresponding to 180 seconds of received samples. All windows produced valid AoA estimates, with a mean corrected correlation of 0.9740 and a minimum of 0.9699. No rejected windows or AoA estimation errors were recorded.

The reported AoA had a mean of 0.6775° and standard deviation of 0.0381°, while the residual phase had a mean of −2.1285° and standard deviation of 0.1195°. Figure 12 shows the continuous AoA output throughout the experiment.

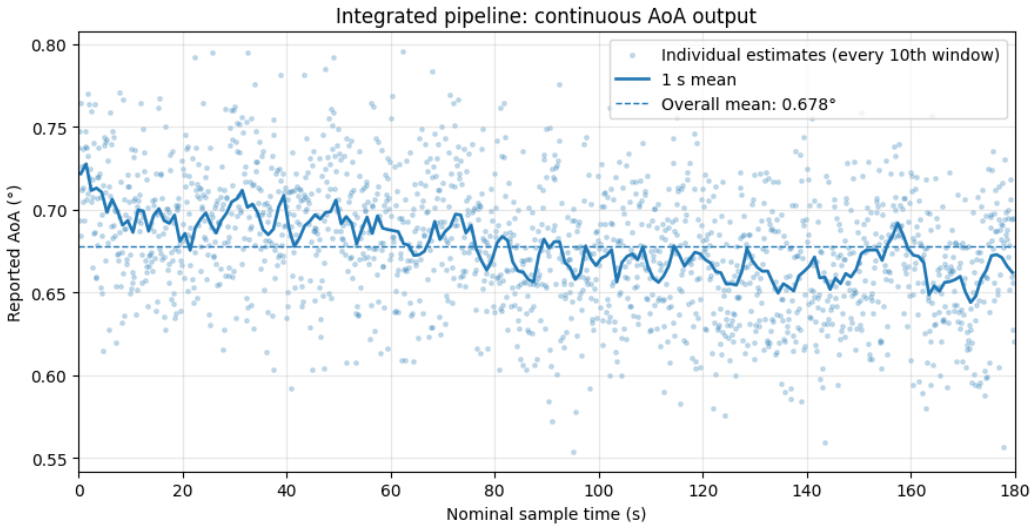


Figure 12: Reported AoA angle over time for a 180 second trial.

These results demonstrate successful integration of RF switching, continuous dual-SDR acquisition, calibration correction and phase-based AoA estimation. All processing windows produced valid estimates, with low variation in the reported AoA throughout the experiment. However, the measured mean angle of 0.6775° represents an apparent AoA derived from the residual interchannel phase, rather than a verified physical arrival direction. As both receivers were supplied with the same conducted RF source rather than spatially separated antennas, this experiment validates pipeline operation and output stability, but not absolute angular accuracy. Hardware validation using a signal with a known physical arrival direction remains outstanding.

6. Discussion

6.1 Synchronisation and Calibration Performance

The results demonstrate that the two bladeRF receivers can be synchronised using a shared reference clock and calibrated to compensate for relative time-delay, amplitude and phase differences. The initial calibration algorithm was validated in simulation and hardware, while improvements to fractional-delay estimation, candidate-delay optimisation and frequency masking increased the reliability and adaptability of the calibration process.

Continuous acquisition experiments demonstrated that a fixed calibration could maintain effective sample alignment over extended periods, with high corrected correlation and no detected residual integer-delay errors. However, gradual changes in residual phase and amplitude were observed despite the calibration parameters remaining unchanged. These changes may be associated with thermal drift, although temperature was not monitored therefore the cause cannot be confirmed. Processing performance was generally sufficient to keep pace with the incoming data, with occasional longer processing intervals accommodated by receiver buffering.

6.2 Calibration Transfer and Phase Repeatability

The most significant limitation identified was the inconsistent residual phase offset when transferring calibration from NOISE to ANTENNA mode. While differences between the two RF paths are expected to introduce a phase offset, this would ideally remain consistent between experiments under unchanged operating conditions. Instead, the initial ten-start investigation produced a total phase-offset spread of 15.084°, despite maintaining high corrected correlation and stable sample alignment.

Further testing demonstrated that this variation could not be attributed solely to RF switching, master transmitter shutdown or the slave transmitter used to generate the ANTENNA-mode signal. Replacing the slave transmitter with an independent noise source allowed transmission to be disabled during switching, but the phase variation persisted, with an 8.635° spread across five independent starts. Frequency-domain analysis also indicated that substantial residual fractional delay was unlikely to explain the observed offsets.

Using two receiver channels on a single bladeRF produced more repeatable phase measurements in the limited tests performed. Although this suggested that the dual-SDR configuration may contribute to the variation, the experiment removed several potential sources of error simultaneously and therefore did not isolate the underlying cause. Further testing of this configuration was not pursued, as it differed from the intended dual-SDR architecture and provided limited scope for identifying the specific source of variation. The underlying cause therefore remains unresolved.

Although the startup-dependent phase offset remained unresolved, the system demonstrated relatively stable phase measurements within individual acquisitions. The external noise-source experiments recorded within-run residual phase standard deviations of 0.067°–0.193°, with no detected residual integer-delay errors across 30,000 processing windows. Gradual phase drift was also observed in earlier stability testing, including a change of approximately +0.20° over 180 seconds, potentially due to thermal effects, although this was not confirmed. These findings suggest that if the additional phase offset could be independently measured and compensated for, the existing system could provide a promising foundation for accurate AoA estimation, subject to physical validation.

6.3 Implications for AoA Estimation and Overall System Performance

Simulation results demonstrated that the phase-based AoA estimator could accurately recover known arrival angles under ideal calibration conditions, with accuracy improving as SNR and sample count increased. However, introducing unknown residual phase offsets substantially reduced angular accuracy. At 30° AoA, a simulated ±7.5° phase-offset distribution increased RMSE to 1.477°, compared with approximately 0.1° under ideal calibration. This demonstrates that the calibration-transfer uncertainty could dominate the AoA error even when the underlying estimator performs accurately.

A simulated secondary known-angle reference substantially reduced this error, indicating that additional phase calibration could address the limitation. However, this method was not implemented or validated on the physical system.

The final integrated experiment demonstrated successful operation of the complete calibration, RF switching, continuous correction and AoA processing pipeline across 18,000 consecutive processing windows. The reported AoA remained stable, but the test used a common conducted signal rather than spatially separated antennas. It therefore demonstrated operational integration and output stability, not physical angular accuracy.

Overall, the project successfully developed and integrated a complete dual-SDR AoA estimation pipeline, incorporating synchronisation, calibration, RF switching, continuous correction and phase-based angle estimation. The system demonstrated reliable operation and stable outputs during continuous acquisition, with the primary remaining limitation being the unresolved startup-dependent phase offset between calibration and measurement. Resolving this additional phase uncertainty would allow the existing pipeline to be evaluated for accurate physical AoA estimation, with known-angle hardware testing then required to validate its performance.

7. Remaining Limitations and Future Work

1. **Investigate the startup-dependent phase offset**
    - Isolate the source of the inconsistent NOISE-to-ANTENNA phase offset through controlled testing of the independent SDR receiver chains, local oscillators and RF signal paths.
    - Repeat measurements across independent starts to establish phase-offset repeatability and identify contributing factors.
2. **Implement a secondary phase calibration**
    - Develop a method of measuring and compensating for the residual phase offset after switching to ANTENNA mode.
    - Investigate the feasibility of using a known-angle reference signal or an additional calibration path without interrupting receiver synchronisation.
3. **Validate AoA using physical antennas**
    - Replace the common conducted signal with spatially separated antennas and a transmitter at known arrival angles.
    - Measure angular error across different angles, SNR levels and operating conditions, comparing hardware performance against simulation.
4. **Characterise long-term phase stability**
    - Investigate possible thermal effects by monitoring receiver temperature and residual phase over extended acquisitions.
    - Determine whether periodic phase correction is necessary to maintain angular accuracy.
5. **Evaluate system robustness and real-time performance**
    - Test repeated calibration and switching cycles using the source-off procedure, including recovery from unsuccessful calibration attempts.
    - Verify processing continuity, sample loss and receiver queue behaviour during extended operation.

8. Conclusion

This project developed and validated a dual-bladeRF software-defined radio system for phase-based angle-of-arrival estimation. The complete processing pipeline was implemented, integrating receiver synchronisation, calibration, RF switching, continuous signal correction and AoA estimation.

Experimental testing demonstrated effective alignment and calibration of the two receivers, with corrected correlations typically exceeding 0.99 during calibration validation. The final integrated system successfully processed 18,000 consecutive measurement windows, producing stable AoA estimates throughout the experiment. Simulation testing also demonstrated accurate angular estimation under ideal calibration conditions, with performance improving as SNR and sample count increased.

The majority of the project requirements were successfully met, with three remaining partially satisfied or requiring further validation. Hardware-induced offsets could be compensated for during calibration (R3.2), although inconsistent residual phase offsets remained between independent starts. Calibration stability was characterised through several experiments (R4.6), but further testing under different operating conditions is required. The AoA accuracy requirement (R6) was demonstrated through simulation, but could not be validated experimentally using known arrival angles.

The principal limitation was the inconsistent residual phase offset between NOISE-mode calibration and ANTENNA-mode measurements. Although phase measurements remained stable within individual acquisitions, the offset varied between independent system starts. Further investigations narrowed the potential causes but did not resolve the inconsistency.

The project established a functional and integrated foundation for dual-SDR AoA estimation. Future work should focus on resolving the residual phase offset, further characterising calibration stability under different operating conditions, and conducting physical testing at known arrival angles to establish the system's absolute angular accuracy.

9. References

[1] P. Delos, B. Broughton, and J. Kraft, “Phased Array Antenna Patterns—Part 2: Grating Lobes and Beam Squint,” *Analog Devices*. https://www.analog.com/en/resources/analog-dialogue/articles/phased-array-antenna-patterns-part2.html

[2] M. Nentwig, “Delay estimation by FFT,” Markus Nentwig, Sep. 22, 2007. https://www.dsprelated.com/showarticle/26.php

[3] Nuand LLC, “bladeRF 2.0 micro - Nuand,” *Nuand*. https://www.nuand.com/bladerf-2-0-micro/#blade-featured-wapper

[4] Skyworks Solutions, Inc., “SKY13330-397LF: 0.1 to 6.0 GHz SPDT Switch,” Data Sheet, Doc. 201407E, Mar. 30, 2020. [Online]. Available: https://www.skyworksinc.com/-/media/SkyWorks/Documents/Products/601-700/SKY13330_397LF_201407E.pdf

[5] “Getting started: Verifying basic device operation,” *GitHub*. https://github.com/Nuand/bladeRF/wiki/Getting-Started:-Verifying-Basic-Device-Operation

Appendix A – Full Evaluation Against Requirements

Stage 1 - SDR synchronisation/phase coherence

Number	Requirement	Status	Results Section
R1	The system shall receive RF signals from two synchronized BladeRF transceivers.	Met	5.2, 5.8
R2	The system shall characterise the relative differences between the two SDR receiver channels.	Met	5.2-5.4
R2.1	The system estimates the relative amplitude difference between the two SDR receiver channels.	Met	5.1, 5.4
R2.2	The system estimates the relative phase difference between the two SDR receiver channels.	Met	5.1, 5.4
R2.3	The system estimates the relative time delay between the two SDR receiver channels.	Met	5.1, 5.4
R3	The system shall calibrate the SDR receiver channels to compensate for hardware-induced offsets.	Met	5.2-5.6
R3.1	The system shall estimate the hardware-induced channel offsets using a common reference signal.	Met	5.3, 5.4
R3.2	The system shall compensate for hardware-induced offsets between the SDR receiver channels.	Partially Met	5.2, 5.6
R4	The system shall validate the performance and stability of the SDR calibration.	Met	5.1, 5.6
R4.1	The system shall evaluate the accuracy of the amplitude, phase, and time-delay estimates.	Met	5.1
R4.2	The system shall characterise the effect of SNR on estimation accuracy.	Met	5.1
R4.3	The system shall characterise the effect of sample count on estimation accuracy.	Met	5.1
R4.5	The system should characterise the temporal stability of the channel calibration.	Met	5.1, 5.2
R4.6	The system shall characterise the stability of the channel calibration under different operating conditions.	Partially Met	5.4-5.6
Stage 2 - Estimating Angle of Arrival			
R5	The system calculates the angle of arrival of the received waveform	Met	5.7, 5.8
R6	The angle of arrival should be calculated to within a determined tolerance.	Simulation only	5.7